

Qatar-1b

R-2774

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120713 · created 2026-09-03T23:28:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456121.92103 +/- 0.00017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.68 +/- 2.11 [%]
Orbital Inclination (inc)	88.29 +/- 2.85
Airmass coefficient 1 (a1)	41.98 +/- 23.42
Airmass coefficient 2 (a2)	-2.98 +/- 1.09
Scatter in the residuals of the lightcurve fit is	48.18 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	3.75
Transit Duration (day)	0.0822 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.1463 (deviation 0.83σ)
Duration fit vs published	0.0822 d vs 0.0692 d (deviation 1.52σ)
Mid-time O–C	-0.2 min
Depth SNR	3.5
Residual scatter	48.18 %

Light curve

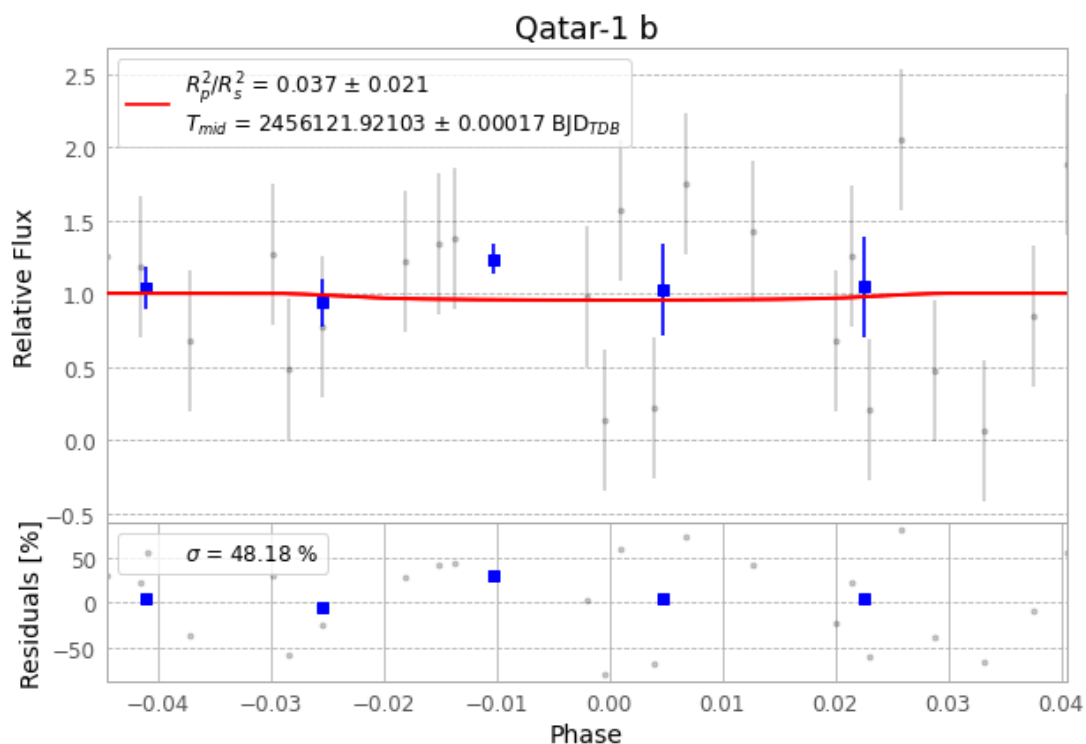


Figure 1 — FinalLightCurve_Qatar-1 b_2012-07-13.png (SHA-256 508522000553a7ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [589, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52fe933856e46144...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-1120713083312.FITS → Qatar-1120713112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120713.log (681f1607a1d0...), FinalLightCurve_Qatar-1 b_2012-07-13.png (508522000553...), FinalLightCurve_Qatar-1 b_2012-07-13.csv (5e71db6c7131...)

Compute resources

Wall time 325.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:28Z.